

WestQuant Representation Scheduler (Project A)

Slide 1 — Project Overview

Problem: A fixed logical Hamiltonian does not determine a unique useful quantum representation. The physical embedding/layout matters.

Approach: Automated search over QoolQit embeddings (InteractionEmbedder, SpringLayoutEmbedder, Blade) using successive halving across 5 stages: cheap metrics, logical fidelity, compilation, emulation, robustness.

Results: On a 5-node MWIS instance, different embeddings of the same Hamiltonian yield solution probabilities from 2% to 11%. The best embedding outperforms the default by >2x.

Importance: Treating the embedding as an optimization variable (not a fixed choice) materially improves quantum solution quality. Open-source, deterministic, no proprietary infrastructure required.

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Slide 2 — QoolQit Experience

Positive: QoolQit's Register, Drive, QuantumProgram abstractions are clean and composable. The embedding API (InteractionEmbedder, Blade) is well-designed. `compile_to(device)` handles device constraints automatically. LocalEmulator with QutipBackendV2 works out of the box.

Challenges: DMM drives require AnalogDeviceWithDMM (not AnalogDevice). The `max_energy` profile is needed for automatic rescaling. Bitstring results come as Counters of strings, requiring careful bit-ordering handling.

Workflow: Successive halving (cheap metrics first, expensive emulation last) is essential for search scalability. QoolQit's `interaction_matrix()` and $1/r^6$ convention made interaction-fidelity metrics straightforward.